Economic complexity - Extra homework 2

1. Data

I used the country-product export data from the Atlas of Economic Complexity (Harvard Growth Lab). The file has exports by country, product and year for the HS 1992 classification at the 4-digit level, from 1995 to 2024. I used the year 2000. In the code, I first removed the codes for unclassified products and the rows with zero exports. Then I built a matrix with countries in the rows, products in the columns and export values in the cells. I also dropped very small exporters, those with less than 0.015% of world exports, because their export baskets are too small to say much about. This left 119 countries and about 1,240 products.

2. Constructing the relatedness network

To build the relatedness network I computed RCA (revealed comparative advantage) as in Hidalgo et al. (2007). A country has RCA in a product if the product's share in its exports is at least as large as the product's share in world exports. I turned this into a 0/1 matrix, where 1 means the country exports the product with RCA ≥ 1, and kept only products with at least 3 such exporters, because results for products with one or two exporters are not reliable.

For every pair of products I counted how many countries export both with RCA ≥ 1. The proximity of two products is this number divided by the larger of the two products' numbers of exporters, which equals the smaller of the two conditional probabilities in Hidalgo et al. The result is a symmetric 1,240 × 1,240 matrix with values between 0 and 1.

The typical product is exported with RCA by about 17 countries. Of the roughly 768,000 possible product pairs, about 727,500 (95%) have a proximity above zero, so almost every product is linked to almost every other, but most links are weak. Among linked pairs the median proximity is 0.18, and only about 0.3% of the pairs are above 0.55.

3. Visualizing the network

Because almost every product is linked to almost every other, the full network cannot be drawn, so I filtered the links as in Hidalgo et al. I kept the maximum spanning tree, which connects all products using the strongest possible links, and added all links with proximity of at least 0.55. The result is one connected network with about 2,700 links. Nodes are colored by product category (I grouped the HS sections into 9 groups so the legend is readable), and node size shows the product's share of world exports. In the force-directed layout, strongly linked products end up close together.

Truong Ngoc Linh (HZU2HX)

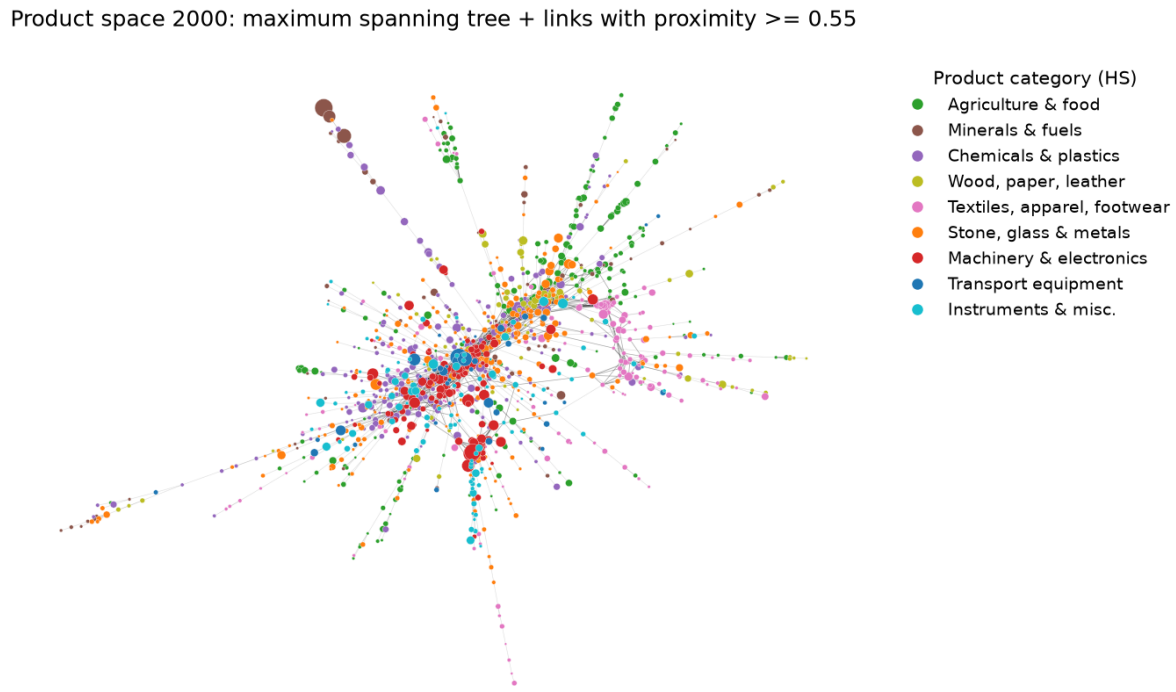


Figure. Product space for 2000. Nodes are 4-digit HS1992 products; two products are linked when the same countries export both with RCA >= 1.
Source: Harvard Growth Lab Atlas (HS92).

1. Figure Product Space for 2000

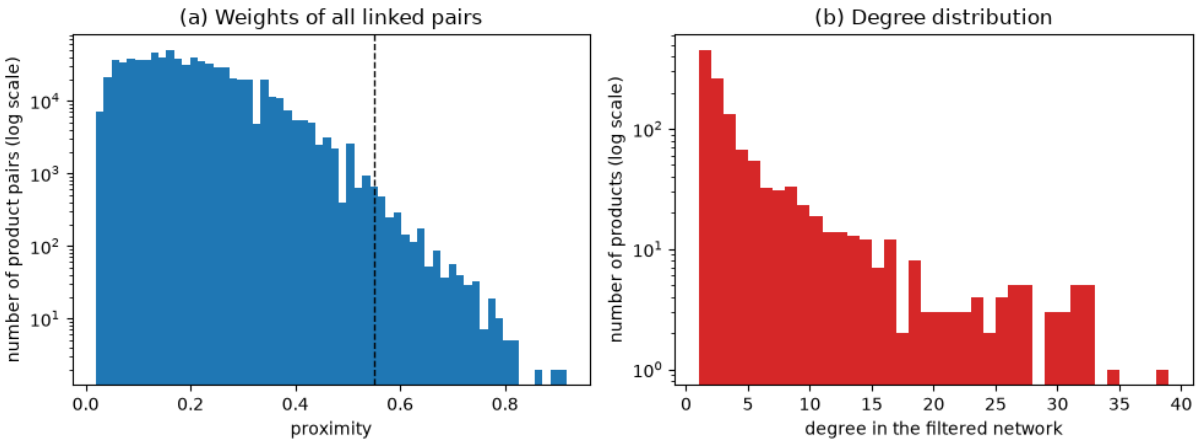
4. Describing the network

Weights: Most proximities are small. The distribution is skewed to the right, with a long thin tail (Figure 2a), and the filter keeps only the top of that tail.

Degrees: Most products have few links (median 2, mean 4.4), and about 36% have only one (Figure 2b). A few hubs have up to 38 links. They are everyday goods such as coated paper, rubber, garments and footwear, which many countries export together with many other products.

Communities: The Louvain algorithm found 37 communities (modularity 0.77). They only partly match the HS sections: using the sections as groups gives a modularity of only 0.33, and only 43% of links connect products from the same section. The textile and apparel community is the clearest (the pink cluster). Machinery, chemicals and metals are split over several mixed communities in the center. Products are related because the same countries can make them, not because they share an official category.

Core and periphery: The innermost k-core has 24 products, almost all garments and footwear, with about 2% of world exports. Countries that export one of them tend to export nearly all. The periphery is large, and many products hang on the network by a single spanning-tree link (the long chains in the figure). The core is not the most sophisticated part of the network: garments have low complexity, and the correlation between centrality and complexity is weak (about 0.2).

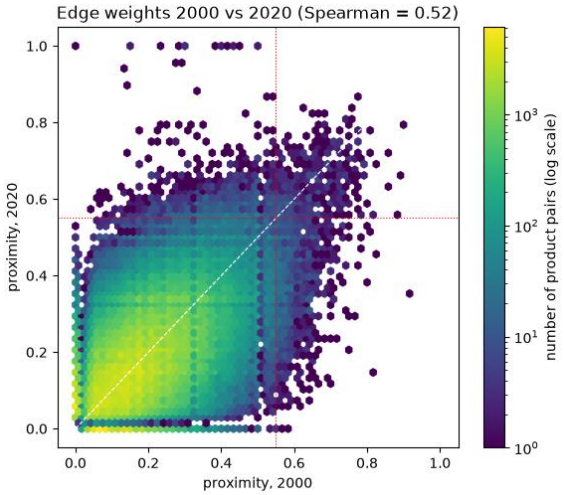


2. Figure (a) Proximity of all product pairs with proximity above zero, 2000 (log scale). (b) Degree distribution in the filtered network (log scale).

5. Comparing 2000 and 2020

I built the proximity network for 2000 and 2020 with the same method, using the products that exist in both years, and compared the proximity of every product pair across the two years (Figure 3). The Spearman correlation is 0.52. The two networks are clearly related, but the points are widely scattered around the diagonal, and many pairs linked in 2000 have almost no link in 2020 (the band along the bottom of the figure).

Strong links persist only partly. Of the pairs with a proximity of at least 0.55 in 2000, about 24% are still above 0.55 in 2020. Nearly all of them stay positive, and their average proximity (0.44) is still well above that of an average pair (about 0.18). Few pairs are strong in both years, which shows in the figure as a thin upper-right corner. The filtered networks (spanning tree plus strong links) share only about 12% of their links (Jaccard similarity). Part of the drop is probably regression to the mean, because the links that look strongest in one year are partly strong by chance. Part of it is also noise: even 2000 and 2001 only correlate at about 0.82. So which products are related is fairly stable, but the exact list of the strongest links changes over time.



3. Figure Proximity of product pairs in 2000 and 2020 (Spearman correlation 0.52)